

NATIVE ELEMENTS

Of the many natural elements, only about 20 are found in the native state; that is, uncombined with other elements. They are divided into three groups. The metals rarely form as distinct crystals, tend to have a high specific gravity, and are relatively soft. Semimetals, such as antimony and arsenic, commonly occur as rounded masses. Nonmetals, including sulfur and carbon, often form as crystals.



NATIVE
COPPER

COPPER

Cubic • H 2½–3 • SG 8.94

Native copper occurs mainly as irregular masses or branching or wire-like forms. It is most notably associated with basaltic lava. A good conductor, it is widely used in the electrical industry.

COPPER ON
GOETHITE
GROUNDMASS



dendritic habit

distinct, isolated
diamond crystal



DIAMOND

Cubic • H 10 • SG 3.51

The hardest of all minerals, diamond, a valuable form of carbon, occurs in igneous rocks called kimberlites, which originate in deep volcanic pipes.



resinous luster



ARSENIC

Trigonal • H 3½ • SG 5.72–5.73

Highly poisonous arsenic usually forms as pale grey, rounded masses in hydrothermal veins. If heated, it smells of garlic.

SULFUR

Orthorhombic • H 1½–2½ • SG 2.07

Native sulfur forms striking yellow crystals and powdery crusts around volcanic vents. It is mined for use in sulfuric acid, dyes, insecticides, and fertilizers.



PLATINUM NUGGET

PLATINUM

Cubic • H 4–4½ • SG 21.44

A rare metal, native platinum forms as scales, grains, and nuggets in igneous rocks and is found in alluvial sands. Its high melting point means it is useful in industry, for example in aircraft spark plugs.

uneven surface



PLATINUM



ANTIMONY

Trigonal

H 3–3½ • SG 6.69

This rare semimetal occurs in hydrothermal veins, often with arsenic and silver minerals. The silvery grey masses are coated white when oxidized.

GRAPHITE

Hexagonal • H 1–2 • SG 2.09–2.23

A form of pure carbon common in metamorphic rocks, graphite is dark, soft, and greasy, and makes ideal pencil leads.

